

Degree-Based Topological Indices Are Exactly Redundant on Molecular Data: A Span Certificate and a Target-Aware Screen for New Descriptors

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Abstract

New degree-based topological indices continue to be proposed at a high rate, and their novelty is typically supported by correlations with properties and with a few existing indices. We show that for the dominant class — bond-incident-degree (BID) indices $\sum_{uv \in E} f(d_u, d_v)$ — this question has an exact, data-free answer on molecular graphs. Building on the known fact, formalised by Rada (2026), that every BID index is a linear combination of the edge-degree-pair counts m_{ij} (ten of them on molecular graphs), we turn it into a dataset-level *span certificate*: a rank condition on the baseline that, when it holds, implies that every BID index, for every f and every parameter value, has zero residual on the baseline, infinite variance inflation and no residual direction that could correlate with any target. A standard 18-index degree-based baseline satisfies the certificate on all graphs of maximum degree at most four, and therefore on all four MoleculeNet benchmarks we study (ESOL, FreeSolv, Lipophilicity, BBBP). A census of 14 published degree-based indices (15 instances, eight of them Sombor-type), together with 100 points of a three-parameter family evaluated on four datasets, confirms exact redundancy to relative residuals below 2×10^{-12} , while non-degree-based Sombor-type controls are correctly not certified. The same analysis shows that partial correlations reported for such indices with absolute numerical tolerances — including those in an earlier draft of this work — are floating-point artefacts. For indices outside the BID class we give a target-aware screen with confidence intervals and a cross-fitted non-linear check: of 20 alternative candidates, 15–17 per dataset pass the pairwise screen, but only two candidate–dataset pairs retain residual target association under both linear and non-linear residualization. Finally we quantify an information ceiling common to all hydrogen-suppressed, unlabelled-graph descriptors: many molecules share a graph, and the target spread within such groups suggests a ceiling on what any topological index, or any model built only from them, can explain.

Keywords. Topological index; degree-based index; Sombor index; redundancy; partial correlation; QSPR; molecular descriptors.

Code and data. <https://github.com/Singati2/topological-index-orthogonality>

1 Introduction

Scalar topological indices have grown from a handful of early proposals [1, 2, 3] to several hundred, and the proliferation has been criticised within mathematical chemistry itself [4]. The growth is fastest for degree-based indices: since the Sombor index was introduced in 2021 [5],

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modified, p -, KG-, elliptic, Euler, diminished and hyperbolic Sombor variants have followed [6, 7, 8, 9, 10, 11, 12]. A new index is commonly supported by a correlation with a property on a small isomer set, by bounds and extremal graphs, and by its correlations with existing indices, e.g. intercorrelation tables among Sombor-type variants [25, 9]. Quality criteria that go beyond single correlations exist — structure sensitivity and abruptness [13], applicability tests on octanes [14], and clustering-based usefulness scores [15, 16] — but none of them answers the question a QSPR modeller needs answered: *does the new index carry information about a target that an explicit baseline does not already carry?*

For the largest class of indices this question turns out to be settled before any data are examined. It is classical that a bond-incident-degree (BID) index $I_f(G) = \sum_{uv \in E(G)} f(d_u, d_v)$ is the linear combination $\sum_{i \leq j} f(i, j) m_{ij}(G)$ of the edge-degree-pair counts m_{ij} [4, 17], and the resulting finite-dimensional linear view of vertex-degree-based indices has recently been developed in general [18, 19], including the observation that the analytic form of f is irrelevant once its values on the degree pairs are fixed. That framework contains no statistical or dataset-level analysis; we draw its consequence for redundancy screening. If a baseline of classical degree-based indices spans the edge-degree-pair counts realised in a dataset — a rank condition that does not involve the candidate or the target — then *every* BID index, whatever its functional form or parameters, is an exact linear combination of the baseline on that dataset, and a target-aware redundancy screen has nothing to measure. We call this rank condition a *span certificate*: a baseline is *certified* on a dataset when the condition holds, and a candidate is then *certified redundant*.

Contributions.

- (1) *Span certificate* (Section 4). The mathematics is not ours: it is the linear framework of Rada [19] and earlier work. We state its statistical consequence for redundancy screening as an explicit, checkable rank condition under which every BID index has zero residual, infinite variance inflation, and partial correlation exactly zero with every target; give the corresponding data-free condition for graphs of maximum degree at most four; and show that a standard 18-index degree-based baseline satisfies it.
- (2) *Census* (Section 5). Fourteen published degree-based indices (15 instances, eight of them Sombor-type) and 100 points of a three-parameter family, on four MoleculeNet datasets: all exactly redundant, which sharpens earlier correlational reports of near-redundancy into an exact statement; two non-BID Sombor-type controls correctly fall outside the span.
- (3) *A numerical pitfall* (Section 6). For exactly redundant indices, partial correlations computed with absolute residual tolerances are noise of order 10^{-2} ; we give a numerically safe recipe.
- (4) *Target-aware screen for non-BID indices* (Section 7). Confidence intervals against an effect-size margin, plus a cross-fitted non-linear residualization, on 20 alternative candidates.
- (5) *Information ceiling* (Section 8). Graph collisions under hydrogen suppression bound what any topological index can achieve; a graph-grouped, corrected benchmark places the indices against an atom- and bond-aware descriptor reference.

The paper is a companion to a mathematical study of the three-parameter Loyola family [20], which concludes on independent grounds that its imbalance parameter is a parsimonious, interpretable direction rather than a predictive upgrade; the span certificate explains why no BID family can add linearly to a classical baseline. (The two papers share authors; the companion paper is in preparation.)

2 Related work

Redundancy and orthogonalization. Strong mutual correlation among topological indices is long established. Hollas studied the redundancy of graph invariants used in chemoinformatics [21] and related the covariance of degree-based indices to the edge structure of the graph [22]; orthogonalization of descriptors is an equally old remedy [23, 24]. Recent work continues to report intercorrelations among new variants, e.g. among Sombor-type indices on octanes [9, 25], and large-scale descriptor evaluations cluster descriptors without reference to a target [15, 16]. We do not claim to discover that topological indices are redundant.

Linear structure of degree-based indices. The identity $I_f = \sum f(i, j)m_{ij}$ is standard [4, 17] and underlies the bond-additive modelling programme in which the term “bond incident degree” index originates [26, 27]. Matrix and finite-dimensional linear formulations of vertex-degree-based indices, including their discrimination properties, are developed in [18, 19]; the M-polynomial [28] is the generating function of the same counts; related algebraic relations between indices appear in [29], and a geometric description of chemical graphs of bounded degree through their edge-type counts in [30, 31]. What we add is not the identity but its statistical consequence for redundancy screening against a baseline and a target: a rank certificate, a data-free version for the molecular degree range, its verification on benchmark data, and a census of published indices.

Degeneracy. That distinct graphs can share index values (degeneracy) and that hydrogen suppression merges distinct molecules are classical concerns [32, 33, 34]; we are not aware of a quantification of graph collisions on the MoleculeNet benchmarks, which Section 8 provides as a ceiling for the whole topology-only representation rather than for a single index.

Descriptor evaluation in QSPR. Topological descriptors remain competitive baselines in modern property prediction [35, 36], which keeps the question of which indices are worth adding practically relevant. Generic feature selection (LASSO [37], mRMR [38], Boruta [39]) acts on a fixed descriptor matrix; the screen here acts one layer earlier, on whether a proposed index should enter that matrix at all.

3 Data, curation and baseline

We use four MoleculeNet benchmarks [40]: ESOL (aqueous solubility) [41], FreeSolv (experimental hydration free energy; column `expt`) [42], Lipophilicity ($\log D_{7.4}$) [43] and BBBP (blood–brain barrier penetration, binary) [44]. SMILES are parsed with RDKit [45]; the largest fragment is kept, the hydrogen-suppressed, atom- and bond-unlabelled graph is formed, and molecules with fewer than two heavy atoms are dropped, so that every graph is connected and has no isolated vertex. On the files as distributed this leaves 1127, 639, 4200 and 2039 molecules. Table 1 reports the counts. Stereochemistry and atom identity are invisible to every descriptor studied here, so molecules that differ only in these respects become duplicates; we report all screening results on the files as distributed and verify in Section 7 that de-duplication on canonical SMILES (averaging regression duplicates whose labels agree within 0.5 target units, dropping duplicate groups that disagree by more or carry conflicting BBBP labels) changes no conclusion; the benchmark of Section 8 uses the curated data. All molecules have maximum degree at most four after hydrogen suppression.

Baseline. The 30-index baseline (Table 2) contains 18 degree-based indices, 8 distance-based indices, 3 spectral indices and the Balaban index. All 18 degree-based indices are BID indices in the sense of Section 4 (vertex-degree sums $\sum_v g(d_v)$ are BID with $f(a, b) = g(a)/a + g(b)/b$

Table 1: Datasets and curation. Raw rows of the MoleculeNet files; invalid SMILES; multi-fragment SMILES reduced to the largest fragment; rows in exact canonical-SMILES duplicate groups; duplicate groups dropped for conflicting labels; molecules after curation (also dropping graphs with fewer than two heavy atoms); distinct hydrogen-suppressed unlabelled graphs (Weisfeiler–Lehman hash); and the percentage of curated molecules whose graph is shared with another molecule.

Dataset	Target	raw	invalid	multi-frag.	dup. rows	conflicts	curated	graphs	shared (%)
ESOL	$\log S$ (mol/L)	1128	0	0	22	1	1115	628	55.5
FreeSolv	ΔG_{hyd} (kcal/mol)	642	0	0	0	0	639	236	74.3
Lipophilicity	$\log D_{7.4}$	4200	0	1	0	0	4200	3472	28.2
BBBP	BBB penetration (0/1)	2050	11	105	141	11	1955	1693	22.6

on graphs without isolated vertices). Several baseline members are related by exact identities, which we disclose because they matter for numerical rank: $\sigma = F - 2M_2$ and $\text{red}M_1 = M_1 - 4|E(G)| + |V(G)|$. The index labelled “Schultz” in our code is the degree distance $\sum_{\{u,v\}} (d_u + d_v) d(u,v)$ (the Schultz molecular topological index differs by M_1); PI is the edge Padmakar–Ivan index; the augmented Zagreb summand is set to zero on K_2 edges.

Table 2: The 30-index classical baseline.

Family	Indices
Degree-based, BID (18)	$M_1, M_2, mM_1, mM_2, F, R, \text{SCI}, H, GA$ [46], AG, ABC, ABS, AZI, SO [5], SO_{red}, Alb [47], $\sigma, \text{red}M_1$
Distance-based (8)	$W, WW, HR, \text{degree distance}, \text{Gutman}, \text{Mostar}, Sz, PI$
Spectral (3)	Estrada EE, energy, spectral radius
Other (1)	Balaban J

4 The span certificate

Let $\mathcal{G}_\Delta$ be the class of finite simple graphs without isolated vertices and with maximum degree at most Δ , and $P_\Delta = \{(i,j) : 1 \leq i \leq j \leq \Delta\}$, so $|P_\Delta| = 10$. For $G \in \mathcal{G}_\Delta$ let $m_{ij}(G)$ be the number of edges whose end degrees are $\{i,j\}$. A *BID index* is $I_f(G) = \sum_{uv \in E(G)} f(d_u, d_v)$ for a symmetric $f : \{1, \dots, \Delta\}^2 \rightarrow \mathbb{R}$.

Remark 4.1 (classical). $I_f(G) = \sum_{(i,j) \in P_\Delta} f(i,j) m_{ij}(G)$, because the edge classes $\{uv : \{d_u, d_v\} = \{i,j\}\}$ partition $E(G)$ and f is constant on each [4, 17]. For example, the path P_4 has two edges of type $(1,2)$ and one of type $(2,2)$, so $I_f(P_4) = 2f(1,2) + f(2,2)$ for every f , and the star $K_{1,3}$ has three edges of type $(1,3)$, so $I_f(K_{1,3}) = 3f(1,3)$. On $\mathcal{G}_4$ the ten functionals m_{ij} are linearly independent: the complete bipartite graph $K_{i,j}$ has all ij edges of type (i,j) , so its count vector is a multiple of the (i,j) -th unit vector. Hence the BID indices on $\mathcal{G}_4$ form exactly a ten-dimensional space; on connected graphs with at least three vertices $m_{11} = 0$ and the space is at most nine-dimensional.

Now fix a dataset $D = \{G_1, \dots, G_n\} \subset \mathcal{G}_\Delta$ with realised pair set $P_D \subseteq P_\Delta$, count matrix $M \in \mathbb{R}^{n \times |P_D|}$, and a baseline whose BID block consists of k indices with value matrix $\Phi_B \in \mathbb{R}^{k \times |P_D|}$, $(\Phi_B)_{b,(i,j)} = f_b(i,j)$. The baseline’s BID columns are $B = M\Phi_B^\top \in \mathbb{R}^{n \times k}$. Write $\mathbf{1}$ for the all-ones vector.

Proposition 4.2 (Span certificate). *If $\text{rank}[\mathbf{1}, B] = \text{rank}[\mathbf{1}, M]$, then for every symmetric f the vector $z_f = (I_f(G_t))_{t \leq n}$ lies in $\text{col}[\mathbf{1}, B]$. Consequently, for any baseline matrix X whose columns include B and for every target vector y : (i) the least-squares residual of z_f on $[\mathbf{1}, X]$*

is zero; (ii) if z_f is not constant on D , its variance inflation relative to X is infinite; (iii) the partial correlation $\text{pcor}(z_f, y \mid X)$ — the correlation between the residuals of z_f and of y after least-squares regression on $[\mathbf{1}, X]$ — is undefined ($0/0$): z_f carries no direction beyond X , and we report it as 0 by convention — any nonzero value computed in floating point is an artefact (Section 6); and (iv) adding z_f to X leaves every ordinary least-squares fit (fitted values, residuals, R^2) on D unchanged. Penalised linear fits (ridge, LASSO) can change, but only through the geometry of the penalty, not through any new direction in the data.

Proof. By Remark 4.1, $z_f = Mf_D$ with $f_D = (f(i, j))_{(i, j) \in P_D}$, so $z_f \in \text{col}(M)$. Since $B = M\Phi_B^\top$, $\text{col}[\mathbf{1}, B] \subseteq \text{col}[\mathbf{1}, M]$, and equal ranks give equality. Thus $z_f \in \text{col}[\mathbf{1}, B] \subseteq \text{col}[\mathbf{1}, X]$, which is (i); (ii) and (iii) restate (i), and (iv) holds because an OLS fit depends on the design only through its column space. $\square$

Corollary 4.3 (data-free certificate on $\mathcal{G}_4$). *If $\text{rank } \Phi_B = 10$ over P_4 , then for every symmetric f there is a coefficient vector c , unique up to $\ker \Phi_B^\top$, with $\Phi_B^\top c = f_{P_4}$, and $I_f(G) = \sum_b c_b I_{f_b}(G)$ holds identically on $\mathcal{G}_4$. In particular the conclusions of Proposition 4.2 hold on every dataset of graphs in $\mathcal{G}_4$, without inspecting the data.*

Proof. $\text{rank } \Phi_B = 10$ means the rows f_b span $\mathbb{R}^{P_4}$; then

$$I_f = \sum_{ij} f(i, j) m_{ij} = \sum_{ij} \sum_b c_b f_b(i, j) m_{ij} = \sum_b c_b I_{f_b}.$$

$\square$

The baseline satisfies the certificate. Evaluating each of the 18 degree-based baseline indices on $K_{i,j}$ gives Φ_B exactly (Table 3); its rank over P_4 is 10 (smallest-to-largest singular value ratio 1.2×10^{-4}), so Corollary 4.3 applies. The certificate is generic rather than delicate: 37 322 of the 43 758 ten-element subsets of the 18 indices already have rank 10, for example $\{M_1, M_2, mM_1, mM_2, F, R, \text{SCI}, H, GA, AG\}$. On the data, the numerical ranks of M , B and $[B, M]$ coincide on all four datasets (Table 4).

Remark 4.4 (what the certificate does and does not say). The certificate is a statement about *linear* redundancy. It does not say that a BID index is useless: a parsimonious index can summarise the ten counts well, and a model that is not rotation-invariant (e.g. a tree ensemble) may use a BID index differently from its constituents. It does say that no BID index can add linearly to a certified baseline, so linear redundancy diagnostics — pairwise correlation, VIF, partial correlation, and the gain of an ordinary least-squares model — are decided in advance for this class. The certificate covers edge-additive indices only; it does not cover normalised or averaged indices such as $I_f/|E|$, multiplicative indices $\prod f$, non-linear transforms such as $\log I_f$, products of indices, neighbourhood-degree-sum and other second-order indices, or valence- and heteroatom-weighted degrees. Beyond $\Delta = 4$ (hypervalent atoms, explicit hydrogens, coordination compounds) the space has dimension $\Delta(\Delta + 1)/2$ and the certificate must be checked on the data, which Proposition 4.2 makes a one-line rank computation.

5 Census of published degree-based indices

We screened the 15 instances of 14 published BID indices in Table 5 — eight Sombor-type instances (the p -Sombor index at two values of p) and seven other degree-based indices — and all 100 points of the Loyola grid $(\alpha, \beta, \gamma) \in \{-1, -\frac{1}{2}, 0, \frac{1}{2}, 1\}^2 \times \{0, \frac{1}{2}, 1, 2\}$ of $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp(\gamma |d_u - d_v| / (d_u + d_v))$ [20] on each of the four datasets. As controls outside the BID class we added the Sombor coindex (sum over non-adjacent pairs) and the eccentric Sombor index (eccentricities in place of degrees).

Table 3: The value matrix Φ_B of the 18 degree-based baseline indices on the ten degree pairs $(i, j) \in P_4$: entry $f_b(i, j)$, obtained by evaluating each index on $K_{i,j}$ and dividing by its ij edges. Vertex-degree sums $\sum_v g(d_v)$ are written as $f(a, b) = g(a)/a + g(b)/b$. The matrix has rank 10 (smallest/largest singular value $1.2e - 04$).

Index	$f(a, b)$	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(2, 2)	(2, 3)	(2, 4)	(3, 3)	(3, 4)	(4, 4)
M_1	$a + b$	2	3	4	5	4	5	6	6	7	8
M_2	ab	1	2	3	4	4	6	8	9	12	16
mM_1	$a^{-3} + b^{-3}$	2	1.12	1.04	1.02	0.25	0.162	0.141	0.0741	0.0527	0.0312
mM_2	$1/(ab)$	1	0.5	0.333	0.25	0.25	0.167	0.125	0.111	0.0833	0.0625
F	$a^2 + b^2$	2	5	10	17	8	13	20	18	25	32
R	$(ab)^{-1/2}$	1	0.707	0.577	0.5	0.5	0.408	0.354	0.333	0.289	0.25
SCI	$(a + b)^{-1/2}$	0.707	0.577	0.5	0.447	0.5	0.447	0.408	0.408	0.378	0.354
H	$2/(a + b)$	1	0.667	0.5	0.4	0.5	0.4	0.333	0.333	0.286	0.25
GA	$2\sqrt{ab}/(a + b)$	1	0.943	0.866	0.8	1	0.98	0.943	1	0.99	1
AG	$(a + b)/(2\sqrt{ab})$	1	1.06	1.15	1.25	1	1.02	1.06	1	1.01	1
ABC	$\sqrt{(a + b - 2)/(ab)}$	0	0.707	0.816	0.866	0.707	0.707	0.707	0.667	0.645	0.612
ABS	$\sqrt{(a + b - 2)/(a + b)}$	0	0.577	0.707	0.775	0.707	0.775	0.816	0.816	0.845	0.866
AZI	$(ab/(a + b - 2))^3$ (0 on K_2)	0	8	3.38	2.37	8	8	8	11.4	13.8	19
SO	$\sqrt{a^2 + b^2}$	1.41	2.24	3.16	4.12	2.83	3.61	4.47	4.24	5	5.66
SO _{red}	$\sqrt{(a - 1)^2 + (b - 1)^2}$	0	1	2	3	1.41	2.24	3.16	2.83	3.61	4.24
Alb	$ a - b $	0	1	2	3	0	1	2	0	1	0
σ	$(a - b)^2$	0	1	4	9	0	1	4	0	1	0
redM ₁	$(a - 1)^2/a + (b - 1)^2/b$	0	0.5	1.33	2.25	1	1.83	2.75	2.67	3.58	4.5

Table 4: Span certificate on the four datasets. $|P_D|$: realised degree pairs; ranks are numerical ranks of the column-centred, column-normalised matrices (relative tolerance 10^{-9}). “Resid.” columns give the largest relative residual $\|r\|/\|z - \bar{z}\|$ on the baseline of the ten counts m_{ij} , of the 15 published BID instances of Table 5, and of the 100 Loyola grid points.

Dataset	n	$ P_D $	rk M	rk B	rk $[B, M]$	resid. m_{ij}	resid. published	resid. Loyola
ESOL	1127	10	10	10	10	4.5×10^{-13}	1.9×10^{-13}	6.9×10^{-13}
FreeSolv	639	10	10	10	10	2.1×10^{-12}	5.6×10^{-14}	1.3×10^{-13}
Lipophilicity	4200	9	9	9	9	2.6×10^{-12}	3.5×10^{-13}	1.9×10^{-12}
BBBP	2039	10	10	10	10	1.1×10^{-12}	1.8×10^{-13}	1.3×10^{-12}

Every published BID index lies in the span of the baseline to a relative residual of at most 4×10^{-13} , and every Loyola grid point to at most 2×10^{-12} ; the two controls have relative residuals of 0.006–0.008 (coindex) and 0.06–0.08 (eccentric Sombor), i.e. they are correctly *not* certified. The universal coefficients of Corollary 4.3 reproduce the data-level values of all 15 instances to relative error below 10^{-10} (file `results/bid_universal_identities.csv`). Two of them reduce to familiar identities, $GG_1 = M_1 + M_2$ and $HM = F + 2M_2$; the others have non-sparse coefficient vectors, which is the generic situation and the reason the redundancy is invisible to a pairwise inspection.

The census is a numerical verification of Corollary 4.3, not independent evidence for it. The pairwise screen happens to reject all 15 published instances here (their maximal correlation with a baseline index is 0.98–1.00), and it passes only one of the 400 Loyola evaluations (100 grid points on four datasets) (FreeSolv, $(\alpha, \beta, \gamma) = (-1, -1, 1)$, $\max |r| = 0.943$). The certificate reaches the same verdicts for a different reason: it is exact, needs no threshold, and would apply equally to a variant whose pairwise correlations happened to fall below any threshold.

Table 5: Census of published BID indices ($f(a, b)$ with a, b the end degrees of an edge) and two non-BID Sombor-type controls. Columns 3–6: largest $|r|$ with any of the 30 baseline indices on each dataset (the conventional pairwise screen). Last column: largest relative residual on the baseline over the four datasets. For KG-Sombor $e = a + b - 2$ is the edge degree. Every BID index is in the span to machine precision; the controls are not.

Index	$f(a, b)$	ESOL	FreeSolv	Lipo	BBBP	max resid.
Modified Sombor [6]	$1/\sqrt{a^2 + b^2}$	0.999	0.999	0.999	1.000	8.6×10^{-14}
Elliptic Sombor [7]	$(a + b)\sqrt{a^2 + b^2}$	1.000	0.999	0.999	1.000	4.1×10^{-14}
Euler Sombor [8]	$\sqrt{a^2 + b^2 + ab}$	1.000	1.000	1.000	1.000	6.8×10^{-14}
Diminished Sombor [9]	$\sqrt{a^2 + b^2}/(a + b)$	1.000	1.000	1.000	1.000	4.6×10^{-14}
Hyperbolic Sombor [10]	$\sqrt{a^2 + b^2}/\min(a, b)$	0.987	0.983	0.985	0.990	3.5×10^{-13}
p -Sombor, $p = \frac{1}{2}$ [11]	$(a^{1/2} + b^{1/2})^2$	1.000	1.000	1.000	1.000	5.9×10^{-14}
p -Sombor, $p = 3$ [11]	$(a^3 + b^3)^{1/3}$	1.000	1.000	1.000	1.000	4.6×10^{-14}
KG-Sombor [12]	$\sqrt{a^2 + e^2} + \sqrt{b^2 + e^2}$	0.999	0.999	0.999	0.999	1.1×10^{-13}
Inverse sum indeg [26]	$ab/(a + b)$	0.998	0.997	0.998	0.998	1.2×10^{-13}
Symmetric division deg [26]	$a/b + b/a$	0.993	0.992	0.992	0.994	1.1×10^{-13}
Nirmala [53]	$\sqrt{a + b}$	1.000	1.000	1.000	1.000	4.7×10^{-14}
First inverse Nirmala [54]	$\sqrt{1/a + 1/b}$	0.999	0.999	0.999	0.999	4.9×10^{-14}
Second inverse Nirmala [54]	$\sqrt{ab/(a + b)}$	1.000	0.999	0.999	1.000	1.8×10^{-13}
First Gourava [55]	$a + b + ab$	0.999	0.999	0.999	0.999	4.6×10^{-14}
Hyper-Zagreb [56]	$(a + b)^2$	0.999	0.998	0.998	0.999	4.1×10^{-14}
Sombor coindex (control)	non-BID	0.999	0.998	0.997	0.999	7.7×10^{-3}
Eccentric Sombor (control)	non-BID	0.971	0.976	0.943	0.947	7.5×10^{-2}

6 A numerical pitfall

An earlier draft of this work reported maximal partial correlations of 0.028 (ESOL), 0.036 (FreeSolv) and 0.009 (Lipophilicity) for the Loyola grid. By Proposition 4.2 the true value is zero. The nonzero values arose because a residual was declared zero only when its absolute standard deviation fell below 10^{-12} ; for indices of magnitude 10^2 – 10^4 the least-squares residual of an exactly redundant index is floating-point noise of relative size 10^{-15} – 10^{-12} but absolute size above that tolerance, and correlating this noise with the target residual gives values of order $1/\sqrt{n}$. Partial correlations are the most affected, because they are a ratio of two noise quantities; VIF and incremental R^2 computed in the same way appear as finite but meaningless numbers (very large, respectively very small). We recommend (a) a relative criterion $\|r_z\|/\|z - \bar{z}\| < 10^{-8}$ with a rank-revealing solver, (b) reporting the in-span status explicitly, and (c) for BID candidates, replacing the statistical computation by the certificate. Our implementation (`src/orthogonality.py`, `span_certified_pcor`) does all three; with it, all 400 Loyola evaluations are exactly 0.

7 A target-aware screen for indices outside the BID class

For a candidate z that is not certified, the question is whether it carries information about a target y beyond the baseline X . We report three quantities.

- Pairwise screen*: $\max_b |r(z, X_b)| < 0.95$, kept only for comparability with current practice.
- Linear partial correlation with a confidence interval*: $\text{pcor}(z, y \mid X)$ from OLS residuals on $[1, X]$, with a Fisher- z interval with standard error $1/\sqrt{n - \text{rk } X - 3}$, where $\text{rk } X$ is the numerical rank of X , read against an effect-size margin $\tau = 0.10$ fixed before the analysis: *pass* if the 95% interval for $|\text{pcor}|$ lies above τ , *negligible* if it lies below, *inconclusive* otherwise. The margin is a convention for a “one percent of residual variance” effect, not a significance threshold.

- (c) *Non-linear conditional association*: z and y are each residualized on X by cross-fitted random forests (five folds), and the correlation ρ_{nl} of the two residual vectors is reported with its Fisher- z interval, in the spirit of double machine learning and the generalised covariance measure [48, 49]. Because the forests are only approximate, we calibrate against three references computed in the same way: an exactly redundant BID index, a pure-noise column, and a deliberately leaky column $y + \varepsilon$.

The 20 candidates span information-theoretic, centrality, spectral-hybrid, motif and eccentricity families (Appendix A); none is claimed as new, and none is a BID index. (Harmonic centrality is identically twice the Harary index and is correctly flagged as in span.)

Table 6: Target-aware screen of the 20 non-BID candidates. Top: counts per dataset — pairwise pass ($\max |r| < 0.95$), point-estimate combined pass ($|\text{pcor}| \geq 0.10$ and pairwise pass), interval pass (pairwise pass and 95% interval of $|\text{pcor}|$ above 0.10), inconclusive, negligible, and in span. Bottom: every candidate–dataset pair with $|\text{pcor}| \geq 0.10$ or $|\rho_{\text{nl}}| \geq 0.10$, with 95% Fisher- z intervals.

Dataset		pairwise	point comb.	interval pass	inconcl.	neglig.	in span
ESOL		15	4	0	12	7	1
FreeSolv		15	1	0	15	4	1
Lipophilicity		16	0	0	1	18	1
BBBP		17	2	2	1	16	1

Dataset	Candidate	$\max r $	linear	pcor [95% CI]	verdict	ρ_{nl}	[95% CI]
ESOL	Bonchev_Id	0.992	−0.113	[−0.171, −0.054]	inconclusive	+0.015	[−0.043, +0.074]
ESOL	CutVerts	0.833	−0.118	[−0.176, −0.059]	inconclusive	−0.015	[−0.074, +0.043]
ESOL	FourCyc	0.189	+0.112	[+0.054, +0.170]	inconclusive	+0.001	[−0.058, +0.059]
ESOL	InfoH_deg	0.699	+0.154	[+0.096, +0.211]	inconclusive	+0.142	[+0.085, +0.199]
ESOL	MeanEcc	0.930	−0.109	[−0.167, −0.050]	inconclusive	−0.028	[−0.087, +0.030]
FreeSolv	EE_x_W	0.985	+0.129	[+0.051, +0.206]	inconclusive	+0.027	[−0.050, +0.105]
FreeSolv	InfoH_deg	0.649	−0.061	[−0.140, +0.018]	inconclusive	−0.104	[−0.181, −0.027]
FreeSolv	Triangles	0.126	+0.107	[+0.028, +0.184]	inconclusive	+0.011	[−0.067, +0.089]
BBBP	FourCyc	0.224	−0.220	[−0.261, −0.178]	pass	−0.171	[−0.212, −0.128]
BBBP	Triangles	0.119	+0.225	[+0.183, +0.266]	pass	+0.013	[−0.031, +0.056]

Calibration of ρ_{nl} across the four datasets: exactly redundant BID reference ($\sum (d_u + d_v)^{-1/2}$, i.e. SCI) −0.044 to +0.043; pure noise −0.030 to +0.039; leaky column $y + \varepsilon$ 0.567 to 0.661.

Results. The pairwise screen admits 15, 15, 16 and 17 of the 20 candidates on ESOL, FreeSolv, Lipophilicity and BBBP. Point estimates $|\text{pcor}| \geq 0.10$ would add four ESOL candidates and one FreeSolv candidate, but with intervals none of them clears the margin: they are inconclusive, with intervals straddling 0.10 (Table 6). This is largely a matter of power: the half-width of a 95% interval is about 0.08 at $n = 639$ (FreeSolv) and 0.06 at $n = 1127$ (ESOL), so a pass would require $|\text{pcor}|$ of about 0.18 and 0.16. On Lipophilicity ($n = 4200$) 18 of the 20 are negligible. Only on BBBP do two candidates, the triangle and four-cycle counts, clear the margin with their whole interval (linear-probability partial correlations 0.225 and −0.220).

The non-linear check separates these further. The calibration references behave as intended: the redundant BID reference and the noise column give $|\rho_{\text{nl}}| \leq 0.05$, and the leaky column gives $\rho_{\text{nl}} = 0.57$ – 0.66 on every dataset. Because on Lipophilicity, the largest dataset, even the two null references have intervals that exclude zero, we use the band $[-0.05, 0.05]$ spanned by the null references, not zero, as the reference, and call a candidate–dataset pair a *robust signal* if both its linear and its non-linear 95% intervals lie entirely outside this band. (This rule is stricter than the margin τ in the sense that it requires agreement of two residualizations, and looser in that it compares with 0.05 rather than 0.10.) Exactly two pairs qualify, and both remain

significant after Benjamini–Hochberg adjustment of the 80 permutation tests ($q = 0.018$): the degree entropy InfoH_deg on ESOL ($\rho_{nl} = 0.14$, $[0.08, 0.20]$) and the four-cycle count on BBBP (-0.17 , $[-0.21, -0.13]$). Several linear “signals” vanish once the baseline is allowed to act non-linearly — most strikingly the BBBP triangle count ($0.225 \rightarrow 0.013$) — which means that their linear residual association was information the baseline already carries in non-linear form. Permutation p -values of a distance correlation between the residuals were not calibrated (the exactly redundant reference was rejected on two datasets) and are not used.

De-duplication. Repeating the linear screen on the curated datasets of Table 1 changes $|\text{pcor}|$ by at most 0.009 and changes no verdict.

8 The information ceiling of topology-only descriptors

Every descriptor considered in this paper — all 30 baseline indices, all BID indices, and all 20 alternative candidates — is a function of the hydrogen-suppressed, atom- and bond-unlabelled molecular graph. Molecules with the same graph therefore receive identical descriptor vectors, whatever their atoms, bonds or stereochemistry. We grouped the curated molecules by a Weisfeiler–Lehman hash [50] of this graph (four iterations), and confirmed with an exact isomorphism test that every one of the 841 groups of molecules sharing a hash consists of isomorphic graphs; the grouping is therefore exact.

The collisions are extensive (Table 1): 55.5% of the curated ESOL molecules and 74.3% of the FreeSolv molecules share their graph with at least one other molecule; FreeSolv’s 639 molecules have only 236 distinct graphs, and the largest group contains 24 molecules. Within groups of molecules that share a graph, the pooled standard deviation of the target is 1.29 (ESOL), 3.18 (FreeSolv) and 0.62 (Lipophilicity), against overall standard deviations of 2.09, 3.85 and 1.20; on BBBP, 15.6% of the shared groups contain both labels. Because no topology-only model can separate molecules within a group, this within-group variance cannot be reduced by any such model (it also contains experimental noise, which limits every model). Two estimates bracket the resulting ceiling. If the pooled within-group variance of the shared groups is representative of the whole dataset, the largest fraction of target variance that any function of the unlabelled graph could explain is about $1 - (1.29/2.09)^2 \approx 0.62$ for ESOL, 0.32 for FreeSolv and 0.74 for Lipophilicity; if instead only the observed within-group scatter is counted (zero for molecules with a unique graph), the in-sample ceilings are 0.83, 0.57 and 0.95. The shared groups over-represent small molecules, and on Lipophilicity only 28% of molecules share a graph, so these are estimates rather than bounds; but on FreeSolv both place the ceiling well below what chemistry-aware descriptors achieve, so the weak performance of topological indices there is a property of the representation, not of any particular index.

Benchmark. To place the indices against this ceiling we ran a downstream benchmark on the curated data (Table 7). Folds are grouped by graph so that identical descriptor vectors never appear on both sides of a split; the design is 5×5 repeated cross-validation; standardisation, screening, and the choice of the LASSO penalty all happen inside the training folds; and paired differences are tested with the Nadeau–Bengio variance correction [51], with equivalence assessed by two one-sided tests (TOST) [52] against a margin of 5% of the reference RMSE, or $0.05(\text{AUC}_{\text{ref}} - 0.5)$ for BBBP; the two one-sided tests at level 0.05 correspond to the reported 90% intervals. As a chemistry-aware reference we used a random forest on RDKit two-dimensional descriptors (180 of RDKit’s 208).¹

¹The remaining 28 (EState- and VSA-EState-type descriptors, BalabanJ, BertzCT and Ipc) terminate the process under the RDKit 2022.09 / NumPy 2 combination used here and were excluded. Three of them are topological; their absence can only weaken the reference, which nevertheless leads by 0.37–2.18 RMSE.

Table 7: Downstream benchmark on the curated data: 5×5 repeated cross-validation with folds grouped by graph, all preprocessing and selection inside the training folds. Top: mean RMSE (ESOL, FreeSolv, Lipophilicity) or ROC-AUC (BBBP), with the mean number of active features in parentheses. Bottom: mean paired difference (model minus reference; lower RMSE and higher AUC are better) with Nadeau–Bengio corrected 90% intervals; “eq.” marks equivalence (TOST) within 5% of the reference RMSE or 5% of the reference AUC above 0.5. *On Lipophilicity the combined screen kept no feature in 24 of 25 folds (intercept-only model).

Model	ESOL	FreeSolv	Lipophilicity	BBBP (AUC)
full 30, LASSO / ℓ_1 -logistic	1.535 (16.0)	3.516 (11.6)	1.130 (13.9)	0.808 (27.1)
pairwise-screened, LASSO	1.580 (7.2)	3.515 (5.9)	1.139 (6.9)	0.791 (5.8)
combined-screened, LASSO	1.590 (4.3)	3.519 (3.1)	1.204 (0.0)*	0.791 (4.0)
full 30, random forest	1.479 (30.0)	3.443 (30.0)	1.086 (30.0)	0.831 (30.0)
pairwise-screened, random forest	1.474 (7.5)	3.429 (7.8)	1.079 (8.0)	0.847 (6.0)
ten counts m_{ij} , random forest	1.458 (10.0)	3.499 (10.0)	1.150 (10.0)	0.841 (10.0)
ten counts m_{ij} , LASSO	1.621 (8.8)	3.498 (7.0)	1.135 (8.6)	0.800 (9.0)
RDKit 2D descriptors, random forest	0.674 (159.7)	1.267 (148.9)	0.714 (160.2)	0.935 (172.5)
Comparison	difference [corrected 90% CI]			
pairwise vs full (LASSO)	+0.045 [+0.023, +0.066] eq.	−0.001 [−0.072, +0.071] eq.	+0.009 [+0.004, +0.013] eq.	−0.018 [−0.036, +0.001]
combined vs full (LASSO)	+0.055 [+0.031, +0.079]	+0.004 [−0.078, +0.085] eq.	+0.073 [+0.060, +0.086]	−0.017 [−0.035, +0.001]
pairwise vs full (RF)	−0.006 [−0.032, +0.021] eq.	−0.014 [−0.099, +0.071] eq.	−0.007 [−0.015, +0.000] eq.	+0.016 [+0.004, +0.028]
m_{ij} vs full 30 (RF)	−0.021 [−0.088, +0.046]	+0.056 [−0.187, +0.299]	+0.064 [+0.039, +0.089]	+0.010 [−0.005, +0.025]
RDKit 2D vs full 30 (RF)	−0.805 [−0.892, −0.718]	−2.176 [−2.441, −1.911]	−0.372 [−0.390, −0.354]	+0.105 [+0.084, +0.126]

Four results stand out. (i) The chemistry-aware reference is far ahead of every topology-only model — by 0.81 RMSE on ESOL, 2.18 on FreeSolv, 0.37 on Lipophilicity and 0.10 AUC on BBBP — and on FreeSolv the best topological model (3.43) is close to the within-graph spread of 3.18, as the collision analysis predicts. (ii) In a random forest the ten counts m_{ij} alone are not distinguishable from the full 30-index baseline on ESOL, FreeSolv and BBBP (intervals include zero, although on FreeSolv the interval is wide); only on Lipophilicity do the distance and spectral indices add measurably (+0.064 RMSE for the counts alone). (iii) Restricting the baseline to its pairwise-screened subset (about 6–8 indices) is equivalent within 5% for the random forest on the three regression sets and slightly better on BBBP; for LASSO it is equivalent within the margin on the regression sets, although the cost is distinguishable from zero on ESOL (+0.045) and Lipophilicity (+0.009), and it is not shown equivalent on BBBP. (iv) The combined screen, which also requires $|\text{pcor}| \geq 0.10$, is a diagnostic and not a feature selector: on Lipophilicity it removes every feature in 24 of 25 folds.

9 Discussion

A concrete benchmark for BID indices. Because every BID index lies in $\text{col}[\mathbf{1}, M]$, its in-sample R^2 in a single-index linear model can never exceed the R^2 of ordinary least squares on the ten counts m_{ij} . On our data that bound is 0.42 (ESOL), 0.19 (FreeSolv), 0.12 (Lipophilicity) and 0.26 (BBBP, as a linear probability model), while the best of the 18 baseline indices, the 15 published instances and the 100 Loyola points reaches 0.36, 0.13, 0.05 and 0.16 (`results/bid_r2_bound.csv`). A proposer of a new BID index can therefore report, on any dataset, how close the index comes to this bound; a modeller who wants the degree information should simply use the ten counts.

What a new degree-based index must show. On molecular graphs (maximum degree four) a new BID index cannot be linearly independent of a baseline that includes ten generic classical degree-based indices; this is a theorem, not an empirical finding. Its justification must therefore come from elsewhere: parsimony (one interpretable number that summarises the ten counts well for a given property, as the Loyola study finds for its imbalance parameter [20]), mathematical properties, or behaviour under models that are not linear in the descriptors. We suggest that a proposal of a new BID index report its coefficient vector $(f(i, j))_{(i, j) \in P_4}$,

which places it exactly in the common ten-dimensional space and makes coincidences between differently named indices immediately visible.

What the target-aware screen adds. For indices outside the BID class the question is genuinely empirical, and the answer depends on uncertainty: of the point-estimate “passers” in an earlier version of this analysis, none survives on ESOL or FreeSolv once intervals are reported, and most linear signals do not survive non-linear residualization. A pairwise threshold, which ignores both the multivariate baseline and the target, is not a meaningful novelty criterion.

What topology cannot do. Section 8 shows that a large share of molecules in these benchmarks are indistinguishable to every hydrogen-suppressed, unlabelled-graph descriptor. This is not a defect of any particular index; it bounds the whole class and explains the weak predictive performance of topological indices on targets such as FreeSolv.

10 Limitations

- The certificate concerns linear redundancy; it does not rank BID indices by usefulness, and it is silent on non-linear models except through Section 7.
- Datasets are four MoleculeNet benchmarks with maximum degree four; other chemistries require the data-level rank check of Proposition 4.2.
- The effect-size margin $\tau = 0.10$ is a convention; we report intervals so that readers can apply their own.
- The non-linear check relies on random-forest residualization; its calibration references set an empirical noise floor of about 0.05, and smaller effects are not resolved.
- BBBP partial correlations use a linear probability model.
- The candidate set of Section 7 is illustrative, not exhaustive.

11 Conclusion

For bond-incident-degree indices on molecular graphs, redundancy is not a matter of thresholds: a rank condition on the baseline, met by any ten generic classical degree-based indices, makes every such index an exact linear combination of the baseline, with zero partial correlation with every target. We verified this on four benchmarks and on fourteen published indices (15 instances), including eight recent Sombor-type variants, and showed that statistical diagnostics computed without this knowledge can produce spurious signal. For indices outside the class, a target-aware screen with intervals and a non-linear check is informative, and on these benchmarks it finds little residual signal. Both results sit under an information ceiling set by graph collisions, which limits every topology-only descriptor.

Data and code availability

Code, curated-data scripts, results and figures are available at <https://github.com/Singati2/topological-index-orthogonality>; the snapshot reported here is git tag v2.0-paper2. Datasets are downloaded from the MoleculeNet distribution by `src/load_data.py`. The analyses of this paper are produced by the following scripts in `scripts/`: 70 (certificate, census, identities), 02, 16 and 17 (screens), 73 (non-linear check), 74 (de-duplication), 72 (curation, collisions, benchmark), 76 (dataset verification), 77 (exact isomorphism check, R^2 bound) and

75 (tables). The four datasets are included in `data/`, verified byte-identical to the MoleculeNet distribution.

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A Definitions of the 20 alternative candidate indices

For reproducibility we give the formal definition of each of the twenty candidate indices screened in Section 7. Let $G = (V, E)$ be a simple connected hydrogen-suppressed molecular graph with $n = |V|$ vertices, $m = |E|$ edges, degree sequence $\{d_v\}_{v \in V}$, geodesic distance matrix $D = (d_{uv})$, adjacency matrix A , eccentricity $\varepsilon(v) = \max_u d_{uv}$, and adjacency-spectrum / Laplacian-spectrum eigenvalues $\{\lambda_i\} / \{\mu_i\}$ respectively. All twenty definitions are implemented in `src/novel_candidates.py` of the companion repository.

Information-theoretic (5). With $H(\{c_i\}) = -\sum_i (c_i / \sum_j c_j) \log(c_i / \sum_j c_j)$ the Shannon entropy of a count distribution:

$$\begin{aligned} \text{InfoH_deg}(G) &= H(\{|\{v : d_v = k\}|\}_k), \\ \text{InfoH_dist}(G) &= H(\{|\{(u, v) : d_{uv} = k, u < v\}|\}_k), \\ \text{InfoH_spec}(G) &= -\sum_i p_i \log p_i, \quad p_i = \lambda_i^2 / \sum_j \lambda_j^2, \\ \text{InfoH_ecc}(G) &= H(\{|\{v : \varepsilon(v) = k\}|\}_k), \\ \text{Bonchev_Id}(G) &= W \log_2 W - \sum_{k \geq 1} k d_k \log_2(k d_k), \end{aligned}$$

where in the last line $d_k = |\{(u, v) : d_{uv} = k, u < v\}|$ and $W = \sum_k k d_k$ is the Wiener index; this Bonchev-type form, as implemented, differs from the standard information-theoretic Wiener index $I_D^W = W \log_2 W - \sum_k k d_k \log_2 k$.

Centrality-based (5). With NetworkX centralities (betweenness normalised by $\binom{n-1}{2}$, NetworkX’s default):

$$\begin{aligned} \text{Btw_sum}(G) &= \sum_{v \in V} c_B(v), \quad c_B(v) = \binom{n-1}{2}^{-1} \sum_{s \neq t \neq v} \frac{\sigma_{st}(v)}{\sigma_{st}}, \\ \text{Cls_sum}(G) &= \sum_v (n-1) \left(\sum_{u \neq v} d_{uv} \right)^{-1}, \\ \text{Eig_sum}(G) &= \sum_v x_v, \quad x \text{ the (positive) Perron eigenvector of } A, \\ \text{Harm_cent}(G) &= \sum_v \sum_{u \neq v} \frac{1}{d_{uv}} = 2 \text{HR}(G), \\ \text{Btw_var}(G) &= \text{Var}_v(c_B(v)). \end{aligned}$$

Spectral-hybrid (3). With $\text{diam}(G) = \max_{u,v} d_{uv}$ and Estrada index $\text{EE}(G) = \sum_i e^{\lambda_i}$:

$$\begin{aligned} \text{AlgConn}(G) &= \mu_2(G), \text{ second-smallest Laplacian eigenvalue,} \\ \text{Rho} \times \text{Dia}(G) &= \rho(G) \cdot \text{diam}(G), \quad \rho(G) = \max_i |\lambda_i|, \\ \text{EE} \times \text{W}(G) &= \log(1 + \text{EE}(G)) \cdot \log(1 + W(G)). \end{aligned}$$

Motif / structural (4).

$$\begin{aligned}\text{Triangles}(G) &= \frac{1}{6} \text{tr}(A^3), \\ \text{MeanClust}(G) &= \frac{1}{n} \sum_v \frac{2t(v)}{d_v(d_v - 1)}, \quad t(v) = \text{triangles at } v, \\ \text{FourCyc}(G) &= \frac{1}{8} (\text{tr}(A^4) - 2 \sum_v d_v(d_v - 1) - 2m), \\ \text{CutVerts}(G) &= |\{v \in V : G - v \text{ is disconnected}\}|.\end{aligned}$$

Eccentricity-based (3).

$$\begin{aligned}\text{TotEcc}(G) &= \sum_{v \in V} \varepsilon(v), \\ \text{Radius}(G) &= \min_{v \in V} \varepsilon(v), \\ \text{MeanEcc}(G) &= \frac{1}{n} \sum_{v \in V} \varepsilon(v).\end{aligned}$$

None of these twenty candidates is claimed as a newly invented index. The information-theoretic family follows the information-theoretic descriptor tradition [?]; the centrality family follows the standard graph-theory literature; the motif and eccentricity families are textbook graph invariants; the spectral-hybrid combinations are deliberately constructed novel pairings to test whether non-standard product indices escape the redundancy ceiling. They appear here purely as screening targets to test the generality of the target-aware screen, and the verdict reported in Section 7 is a statement about this particular candidate set under the tested baselines and thresholds, not a claim that any individual candidate is or is not a useful QSPR descriptor.

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